

1. Types of Malware

Malware (malicious software) is designed to harm, exploit, or otherwise compromise a computer system, network, or device.

Common types include:

- **Virus:** Attaches to legitimate files and spreads by infecting other files.
 - **Worm:** Self-replicates across networks without needing host files.
 - **Trojan Horse:** Disguises as legitimate software but performs malicious actions.
 - **Ransomware:** Encrypts data and demands ransom for decryption keys.
 - **Spyware:** Secretly monitors user activities and sends data to attackers.
 - **Adware:** Automatically displays or downloads unwanted ads.
 - **Rootkits:** Conceals presence of malware by hiding system modifications.
 - **Keyloggers:** Records keystrokes to steal passwords and sensitive info.
 - **Botnets:** Networks of infected devices controlled remotely for attacks.
 - **Fileless Malware:** Operates in memory without writing files, evading detection.
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2. Malicious Code Families

Malicious code refers to all types of malware and scripts designed to harm systems.

Families include:

- **Executable Malware:** Viruses, worms, Trojans.
 - **Script-based Malware:** JavaScript, VBScript, PowerShell scripts.
 - **Macro Malware:** Embedded in documents (e.g., Word, Excel macros).
 - **Logic Bombs:** Trigger malicious actions under specific conditions.
 - **Bots:** Malware that turns systems into part of a botnet.
 - **RATs (Remote Access Trojans):** Provide remote control to attackers.
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3. Latest Trends in Malware

- **Fileless Attacks:** Use legitimate tools (PowerShell, WMI) to evade antivirus.
- **Ransomware Evolution:** Targeting large enterprises, demanding multi-million-dollar ransoms.
- **Malware-as-a-Service:** Offering malware tools for rent on dark web.

- **Polymorphic Malware:** Changes code to avoid signature detection.
 - **Mobile Malware:** Increasing attacks on Android/iOS devices.
 - **AI-powered Malware:** Uses AI to improve evasion and targeting.
 - **Supply Chain Attacks:** Infecting software updates or third-party providers.
 - **Cryptojacking:** Unauthorized mining of cryptocurrency on victim devices.
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4. Analysis of Malware

Malware analysis involves examining malware samples to understand functionality, origin, and impact.

Two main types:

- **Static Analysis:** Examining code without execution. Techniques:
 - File inspection (strings, headers)
 - Disassembly and reverse engineering
 - Signature-based detection
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Dynamic Analysis: Running malware in controlled environments (sandboxing) to observe behavior:

- Network traffic monitoring
- System changes tracking (file creation, registry edits)
- API calls tracing

Malware analysis helps develop detection methods and create mitigation strategies.